Shenguanlin Lin

ALGANT Master's student | Université de Bordeaux

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Research interests

Arithmetic geometry and p-adic Hodge theory, with particular interests in perfectoid spaces, Fargues–Fontaine curves.

Education

M.Sc. in Mathematics — ALGANT programme

2025–2027 (expected)

Université de Bordeaux, France & University of Regensburg, Germany

- **2026–2027: Université de Bordeaux.** Second year, M2 AGTN / ALGANT.
- **2025–2026: University of Regensburg.** First year. Academic mentor: Walter Gubler.

Selected courses at Regensburg *German grading scale: 1.0 is the highest grade.*

Non-Archimedean Analytic Geometry	Klaus Künnemann	**1.0**
Local Class Field Theory	Shai Keidar	**1.0**
Introduction to Stable Homotopy Theory	Bastiaan Cnossen	**1.3**

B.Sc. in Mathematics and Applied Mathematics

2021–2025

Southern University of Science and Technology (SUSTech), Shenzhen, China.

GPA: 3.61/4.0. Academic mentor: Hui Gao.

Undergraduate thesis: *On Galois Representations and p-adic Hodge Theory*
An expository study of (φ, Γ)-modules, Fontaine's period rings, and the classification of semistable representations via weakly admissible filtered (φ, N)-modules.

Research experience

Geometric Realizations of Absolute Galois Groups in Perfectoid Geometry

2026–present

Research project; preliminary manuscript in revision.

- Generalized Weinstein's construction to complete rank-one nonarchimedean fields of mixed characteristic admitting Galois extensions with perfectoid completions, including $E = \mathbb{Q}_p\{\{X\}\}$.
- Constructed a pro-étale sheaf Z and proved an equivalence of Galois categories $\mathrm{FEt}(Z) \simeq \mathrm{FEt}(E)$, using Kedlaya's base-change lemma, tilting, and Galois descent.
- Completed a manuscript containing a full proof of the main theorem; currently revising it following discussions with Professor Denis Benois.

Conference presentation

An introduction to p-adic geometry

December 2025

ALGANT in China 20th Anniversary Meeting. 15-minute talk.

Overview of rigid, Berkovich and adic geometry, perfectoid spaces and tilting, with an introduction to Fargues–Fontaine curves and Weinstein's construction.

Scholarships and awards

2026–2027	ALGANT funding, Université de Bordeaux: CAP IA scholarship and internship stipend from the Institut de Mathématiques de Bordeaux (IMB).
2026–2027	Erasmus+ mobility grant for study at Université de Bordeaux through the University of Regensburg.
2025–2026	ALGANT scholarship, University of Regensburg.
2025	Outstanding Undergraduate Graduate, SUSTech.
2022	Third-Class Scholarship for Excellent Students, SUSTech.
2021	Outstanding Freshman Scholarship, SUSTech.

Selected seminar presentations and notes

University of Regensburg, 2025–2026. Expository work prepared for student seminars.

Galois theory in algebra and topology

Coverings of Riemann surfaces, profinite fundamental groups, and Douady's theorem on the absolute Galois group of $\mathbb{C}(t)$.

Normal schemes and the arithmetic fundamental group

Finite étale coverings, unramified extensions of function fields, and the geometric–arithmetic fundamental group exact sequence.

Serre vanishing

Ample and very ample line bundles, Serre's vanishing theorem, and its extension from very ample to ample twists.

Triangulation in o-minimal structures

Definable triangulation, simplicial complexes, and compatibility with finitely many definable subsets.

Selected academic activities

Geometry Summer School, Utrecht University August 2024

SUSTech Mathematics–Physics Workshop June 2024

Languages and tools

Languages: Chinese (native); English (IELTS 7.0).

Tools: LaTeX, SageMath; Python (basic).